

the same time, the figure shows consistent, negative employment effects of the minimum wage for levels far above the minimum wage: indeed, the aggregate negative employment elasticity (e.g., -0.137) accrues almost entirely in wage bins exceeding \$15/hour.

It strikes us as implausible that a minimum wage increase in the \$8 to \$9/hour range causally leads to losses of jobs mostly at or above the median wage, even though the minimum wage is binding far lower in the wage distribution. More plausibly, this suggests that the specification is confounded by negative employment shocks to the upper part of the wage distribution (possibly much earlier than the actual treatment dates), and these shocks are not fully absorbed by the simple two-way fixed effect specifications estimated using a long panel. For instance, as shown in [Online Appendix Table G.2](#), the negative employment changes shown in [Figure VI](#) arise only for the Card and Krueger low-probability group, which should not be affected by the minimum wage. At the same time, the high- and medium-probability groups exhibit no negative disemployment effect.

How is it possible that our benchmark specification that focuses on employment changes around the event window leads to such different results compared to the TWFE-logMW specification? The differences in the estimates suggest that the negative employment changes in the upper part of the wage distribution in TWFE-logMW must come from outside of the event window. Indeed, we find that the employment losses are driven by the 1980s expansion and the 1990–1991 recession, even though most of the minimum wage changes in our sample occurred after 2000. When we restrict the regression to the 1993–2016 period—the period where 86% ($\frac{118}{138}$) of all our events occurred—we indeed find very similar estimates across the two specifications: the TWFE-logMW specification in this sample suggests small employment effects and little upper-tail employment changes ([Online Appendix Figure G.5](#)). Moreover, even if we limit our sample to the 39 states with no state minimum wage increases until the late 1990s, the negative disemployment effects are driven entirely by the inclusion of a period (1980s) long before any cross-state variation in treatment occurred in these states (see [Online Appendix Table G.5](#)).

We also show in [Online Appendix Figure G.4](#) that the overall employment-to-population rate evolved very similarly between the early 1990s and 2016 in high minimum wage states (those that instituted a minimum exceeding the federal standard after

the early 1990s) and low minimum wage states (where the federal standard was always binding). This is noteworthy because the period between the early 1990s and 2016 is when much of the cross-sectional variation in minimum wages emerged. Importantly, however, the employment-to-population rate had diverged between these two groups of states during the 1980s, at least a decade before most high minimum wage states started to raise their minimum wage. This creates a spurious correlation between employment changes in the 1980s and minimum wage changes in the early 2000s that confounds the TWFE-logMW specification in the full sample, which is sensitive to shocks occurring long before the event window. However, this does not affect our event study estimates, since these only consider employment changes within the event window. This also explains why we do not find any pre-existing trends in our event-based analysis, while the TWFE-logMW in the full sample exhibits sizable and statistically significant leads. At the same time, TWFE-logMW estimates restricted to the post-1992 sample produce neither sizable leads nor sizable employment effects (see further details in [Online Appendix G](#)).³⁶

The foregoing example illustrates that showing the effect of the minimum wage throughout the wage distribution can provide additional falsification tests and therefore can be a useful tool for model selection. This type of model selection tool can be particularly helpful in the context of minimum wages, where the literature has often grappled with figuring out the “right” empirical model.³⁷

36. Why are the expansion in the 1980s and the downturn in the 1990–1991 recession related to future minimum wage changes? Because the expansion and downturn were more pronounced for states that would be more Democratic-leaning in the 2000s. One possibility is that the 1990–1991 recession was so severe in some states that it changed the political landscape and opened the door for candidates supporting minimum wages. However, another explanation is that the 1990–1991 recession just happened to be more pronounced for Democratic-leaning states—states that would also be more inclined to raise the minimum wage starting in the early 2000s following a long period of federal inaction. In [Online Appendix G](#) we show that this latter explanation fits the data better. In particular, we show that the predictive power of the severity of recession on future minimum wage increases disappears once we control for the partisan voting index (PVI) in the 2000s and instrument that variable with the PVI in 1988 (i.e., prior to the 1990–1991 recession).

37. [Online Appendix G](#) also uses these findings to provide a reconciliation of other longstanding discrepancies in the minimum wage literature. We show in [Online Appendix Table G.7](#) that the expansion in the 1980s and the downturn dur-

V. DISCUSSION

In this article we infer the employment effects of the minimum wage from the change in the frequency distribution of wages. The key advantage of this approach is that it allows us to assess the overall impact of the minimum wage on low-wage workers, who are the primary target of minimum wage policies. We use an event study analysis exploiting 138 prominent minimum wage increases and provide a robust and comprehensive assessment of how minimum wages affect the frequency distribution of wages. Second, we calculate the number of missing jobs just below the minimum wage, the number of excess jobs at or slightly above the minimum wage, and the job changes in the upper tail of the wage distribution. Our main estimates show that the number of excess jobs at and slightly above the minimum wage closely matches the number of missing jobs just below the minimum wage, and we find no evidence for employment changes at or more than \$4 above the minimum wage. A similar pattern obtains for low-skilled workers, suggesting labor-labor substitution is unlikely to be a factor in our setting. Moreover, we find that the level of the minimum wages that we study—which range between 37% and 59% of the median wage—have yet to reach a point where the job losses become sizable. However, the employment consequences of a minimum wage that surpass the ones studied here remain an open question. Furthermore, if minimum wage increases affect tradeable sectors more, our findings suggest employment effects may be more pronounced.

A key advantage of tracking job changes throughout the wage distribution is that we can transparently show the source of any disemployment effects. As a result, we can detect when an empirical specification suggests an unrealistic impact on the shape of the wage distribution. More important, the relationship between minimum wages and the wage distribution can also be used to infer the structure of low-wage labor markets. While providing a unified theoretical framework is beyond the scope of this article,

ing 1990–1991 is also responsible for the sensitivity of teen employment estimates to specification that has plagued the literature, such as controls for trends (e.g., Neumark, Salas, and Wascher 2014; Allegretto et al. 2017). In the post-1992 period, there is little divergence in teen elasticity across standard specifications (including the TWFE-logMW): they all suggest any losses to teen employment are small, with elasticities no more negative than -0.03 . Use of trend controls also matters little in the post-1992 sample which is where most minimum wage variation is.

our empirical results on the wage distribution together with the estimates on labor-labor substitution across demographic groups and the heterogeneous responses across sectors provide new empirical findings which can be used to test and distinguish various theories of the low-wage labor market.

UNIVERSITY OF MASSACHUSETTS, AMHERST

UNIVERSITY OF MASSACHUSETTS, AMHERST, NATIONAL BUREAU OF ECONOMIC RESEARCH, INSTITUTE OF LABOR ECONOMICS

UNIVERSITY COLLEGE LONDON, CENTER FOR ECONOMIC PERFORMANCE, INSTITUTE FOR FISCAL STUDIES, INSTITUTE OF LABOR ECONOMICS, RESEARCH CENTER FOR ECONOMIC AND REGIONAL STUDIES, HUNGARIAN ACADEMY OF SCIENCES

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SUPPLEMENTARY MATERIAL

An [Online Appendix](#) for this article can be found at *The Quarterly Journal of Economics*. Data and code replicating tables and figures in this article can be found in [Cengiz et al. \(2019\)](#), in the Harvard Dataverse, doi:10.7910/DVN/TJCTC7.

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